

Machine Learning for Climate-Resilient Agriculture

Climate change poses a serious threat to global agriculture. Rising temperatures, changing rainfall patterns and the increased frequency of extreme weather events reduce crop yields and threaten food security in many regions. This thesis investigates how machine learning can support farmers and policymakers in adapting to these challenges.

Despite this progress, the adoption of deep learning in clinical practice remains limited. A key obstacle is the lack of interpretability: clinicians hesitate to trust predictions from models that cannot explain their reasoning.

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Machine learning methods based on layered neural representations have become the dominant artificial intelligence approach of recent years. Such models automatically discover intricate structure in large datasets including medical images, speech and natural language, and they now rival human experts on several diagnostic tasks.

Football analytics is a growing field in sports science. Coaches use tracking data to evaluate player fitness, optimize training loads and reduce injury risk across a long competitive season. Statistical models of passing networks can even quantify a team's playing style.

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We propose an explainable pipeline for predicting mortality in intensive care units, combining gradient boosted trees with SHAP value explanations, and we validate it on the public MIMIC-IV database of critical care records.